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Model-Based Reinforcement Learning Variable Impedance Control for Human-Robot Collaboration

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Abstract

Industry 4.0 is taking *human-robot collaboration* at the center of the production environment. Collaborative robots enhance productivity and flexibility while reducing human's fatigue and the risk of injuries, exploiting advanced control methodologies. However, there is a lack of real-time model-based controllers accounting for the complex human-robot interaction dynamics. With this aim, this paper proposes a *Model-Based Reinforcement Learning* (MBRL) variable impedance controller to assist human operators in collaborative tasks. More in details, an ensemble of Artificial Neural Networks (ANNs) is used to learn a human-robot interaction dynamic model, capturing uncertainties. Such a learned model is kept updated during collaborative tasks execution. In addition, the learned model is used by a *Model Predictive Controller* (MPC) with *Cross-Entropy Method* (CEM). The aim of the MPC+CEM is to online optimize the stiffness and damping *impedance control* parameters minimizing the human effort (i.e, minimizing the human-robot interaction forces). The proposed approach has been validated through an experimental procedure. A lifting task has been considered as the reference validation application (weight of the manipulated part: 10 kg unknown to the robot controller). A KUKA LBR iiwa 14 R820 has been used as a test platform. Qualitative performance (i.e, questionnaire on perceived collaboration) have been evaluated. Achieved results have been compared with previous developed offline model-free optimized controllers and with the robot manual guidance controller. The proposed MBRL variable impedance controller shows improved human-robot collaboration. The proposed controller is capable to actively assist the human in the target task, compensating for the unknown part weight. The human-robot interaction dynamic model has been trained with a few initial experiments (30 initial experiments). In addition, the possibility to keep the learning of the human-robot interaction dynamics active allows accounting for the adaptation of human motor system.

Keywords Human-robot collaboration · Machine learning · Industry 4.0 · Model-based reinforcement learning control · Variable impedance control

1 Introduction

1.1 Human-Robot Collaboration in Industry 4.0

Within the *Industry 4.0* paradigm [1], a new concept of *smart worker* [2] is proposed. In such a scenario, the human operator is in charge of added-value tasks, management of the production process and supervision of the operations, being relieved from onerous tasks. To achieve such an ambitious result, *human-robot collaboration* is a fundamental topic to be investigated [3]. *Physical* human-robot interaction control is widely investigated to empower the operator [4–6]. *Impedance control* [7] is commonly adopted as a low-level control layer [8, 9], implementing a (tunable) mass-spring-damper compliant system. Inspired by bio-mechanical systems, researchers have been exploring the advantages of adapting the impedance behavior of the controlled robot during the task execution [10], i.e, adjusting its stiffness and damping matrices according to a force feedback. In such a way, the controlled manipulator is capable to achieve human-like adaptability skills [11, 12]. In the following, the developed methods in this research field are analyzed.

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1.2 Variable Impedance Control for Human-Robot Collaboration: State of the Art

Two families of controllers can be identified from the state of the art related to variable impedance control for human-robot collaboration purposes: non-learning-based control methodologies and learning-based control methodologies. While the first family of controllers exploits predefined control laws and sensory-data to adapt the robot impedance behavior to collaboration state, the second family exploits sensory-data in order to learn from the human-robot interaction the optimal impedance model based on desired performance. In the following, these two main families of controllers are analyzed.

1.2.1 Non-Learning-Based Control Methodologies

In [13] a variable admittance control method is presented. The method adapts the admittance damping in real-time. Such an adaptation is performed monitoring the co-activation of the antagonist muscles by means of electromyography sensors. In the case that a predefined muscular activation threshold is achieved, the adaptation of the damping parameters is performed. Two assistance modalities are proposed: high-accuracy positioning assistance and low-effort assistance. In [14] an adaptive inverse control approach is developed for the tuning of the admittance model. The aim of the proposed methodology is to achieve a prescribed (known) human-robot dynamics given by an available task model. In [15] a variable admittance controller is proposed. It infers human intentions by monitoring the variation of the interaction force magnitude, accordingly tuning the admittance controller damping. The proposed methodology allows to achieve fine robot positioning (using high damping values) or fast robot motion (using low damping values). In [16] a methodology adapting the admittance parameters to reduce the human effort in the collaboration task is developed. Such methodology exploits energy tanks theory [17] to ensure the passivity of the controller. In [18], the proposed method argued that human may benefit from robot leadership in human-robot co-manipulation tasks. A variable admittance controller is proposed, where stiffness is continuously varied based on a scalar and smooth function that assigns a degree of leadership to the robot. In [19] human morphology is considered, presenting a human arm manipulability-based stiffness adaptation. In [20] an admittance control is proposed. Such methodology generates a reference trajectory for the controlled robot exploiting the interaction force measurements. In addition, the robot dynamics uncertainties are included in the control design by means of an adaptive controller.

1.2.2 Learning-Based Control Methodologies

In [21] a variable impedance control with reinforcement learning algorithm PI^2 is proposed. Such approach is defined by a model-free sampling-based learning method derived from optimal control theory. In [22] a variable admittance control (using the Reinforcement Learning-based Fuzzy Q-Learning FQL algorithm) is proposed. The robot end-effector rotational motion is not considered within the proposed methodology. In [23] an approach that applies probabilistic learning for robot stiffness adaptation is proposed. In such a way, the robot behavior is mapped w.r.t. the task execution. In [24] a model reference adaptive control is proposed. An inner neuro-adaptive controller learns online the robot dynamics to shape a superimposed impedance behavior without the use of the task information. An outer task-specific controller (exploiting a simple pre-defined human operator dynamics) adapts such inner impedance behavior to achieve a target human-robot interaction performance. In [25] an LQR impedance controller is proposed to reduce the human effort. The impedance controller optimization exploits an integral reinforcement learning method to avoid the use of a human dynamics model. In [26] a barrier-Lyapunov-function-based adaptive impedance control is proposed. Such methodology incorporates physical limits, transient perturbations, and time-varying dynamics in the learning procedure. In [27] neural networks to adapt impedance parameters online are proposed. The developed method minimizes the defined cost function, taking into account the trajectory-tracking errors and the measured interaction forces. In [28] a data-efficient learning method for variable impedance control is presented using Gaussian process (GP) model.

1.2.3 Considerations

Even in the case of adaptive controllers, the main limitations of the above mentioned non-learning-based methodologies consist in a limited variation of task conditions that can be faced by the proposed controllers, limiting their applicability. Therefore, such approaches require a good knowledge of the target task dynamics in order to tune the control parameters. In fact, without having the possibility to learn the task dynamics, a complete re-tuning of the controller is not feasible. Learning-based controllers are indeed powerful tools. It is, in fact, possible to exploit such methodologies to automatically tune the task-specific controller through trial-and-error procedures. However, most of the learning-based methods mentioned above are based on a model-free reinforcement learning approach, requiring a too high number of trials (hundreds or even thousands of required experiments) to achieve target performance (not feasible for all industrial applications). Other approaches (such as [24] and the ones related to programming by demonstration and imitation learning [29, 30]) have the focus of the learning only on a specific task objective, without considering the complex human-robot interaction dynamics modeling and learning for model-based control design. Only few of the above described approaches consider such human-robot interaction dynamics modeling. [28] is based on learning the interaction dynamics between the robot and the environment using a GP model. The learned model is then used to optimize the impedance control parameters. Nevertheless, the computation of the new set of impedance control parameters takes minutes, which is far from real-time implementation purposes in industrial environments. In fact, although GP models have high sample-efficiency [31], giving an accurate estimate of model uncertainties (which is essential for model-based approaches to succeed) [32], the main drawback of such approach is that it is very slow when data-set is large (do not scale with the data-set), and it can work only with a certain type of loss functions. Additionally, GP models are not good at representing non-smooth dynamics, typical behavior of high nonlinear systems such as human-robot interaction dynamics.

1.3 Human-Robot Interaction Dynamics Modeling for Control Purposes: State of the Art

Although the human's arm dynamics results to be highly nonlinear and difficult to be modeled [33, 34], human-robot interaction dynamics modeling has been proved to be important in model-based control design for human-robot collaboration, even considering simplified modeling [35–37]. Machine learning techniques have been applied in order to learn such human-robot interaction dynamics to be exploited in model-based controllers. Some of the works exploit *a priori* human's arm dynamics modeling. In [38] a method for human intention estimation using Gaussian Processes (exploiting an empirically identified impedance dynamic model for the human arm) is proposed, allowing the estimation of the human motion intention exploiting the measured interaction forces. In [39] the computation of the interaction forces in a collaborative object manipulation task is proposed. Such method exploits the theoretical studies on the human arm movements to identify a dynamic model. Such dynamics is embedded in the proposed control strategy to perform the collaborative task. However, such approaches are not capable of capture the highly nonlinear behaviors of the human-robot interaction dynamics. Other approaches involve instead the learning of the complex nonlinear human-robot interaction dynamics. In [40] a data-efficient unsupervised reinforcement learning framework is proposed. In such a way, the robot is capable to learn the task from its own sensory-data. The uncertainties related to the human actions is modeled using the Gaussian Processes theory. In [41] a data-driven learning approach is proposed for the control of an exoskeleton robot. By means of electromyography signals, a cost function is proposed to learn the assistive control strategies. In [42] a method for human–robot collaboration is proposed. The controlled robot behavior is adapted online to the human motor fatigue, estimated using a model based on the muscles activity measurements. However, no approaches applying the learned nonlinear human-robot interaction dynamics to variable impedance controller can be found. As highlighted by the above state of the art analysis, the combination of variable impedance control with nonlinear human-robot interaction dynamics modeling may lead to high-performance model-based controllers for human-robot collaboration purposes, making a conceptual step beyond the current available control solutions.

1.4 Paper Contribution

In order to develop an efficient control methodology, capable to model the complex human-robot interaction dynamics and to exploit it for the online optimization of the impedance parameters for human-robot collaboration purposes (i.e, minimizing human-robot interaction forces, therefore, reducing the human effort), a *model-based reinforcement learning* (MBRL) variable impedance control is proposed. In particular, three main components are defined: (i) an impedance controller, (ii) a human-robot interaction dynamics modeling approach, and (iii) an online optimizing impedance control parameters strategy. Considering (i), the impedance control allows designing a compliant behavior for the manipulator, being able to safely interact with the human operator. Considering (ii), the proposed methodology relies on *Artificial Neural Network* (ANN) models [43]. The ANNs are offline trained, learning the human-robot interaction dynamics, and updated periodically, in order to keep the human-robot interaction dynamics learning active (i.e, accounting for the adaptation of human motor system). The obtained model allows correlating the interaction states at time t (i.e, force, derivative of the force, Cartesian position and Cartesian velocity at the robot end-effector) with the control variables (i.e, stiffness and damping impedance control parameters) in order to predict future interaction states at time $t + dt$. In addition, ANNs are very expressive, can represent any nonlinear function (more often present in robotics), and very efficient when the data-set is large (scale well with the data-set using modern GPUs). The main drawback of ANNs is the fact that they suffer from over-fitting when the data-set is small, which leads to poor performance. To overcome such issue, the presented work implements an ensemble of ANNs to learn the human-robot interaction dynamics. In such a way, ANNs have as an output a distribution over a prediction instead of a single value. It is, therefore, possible to capture the uncertainties related to the human-robot interaction dynamics, improving the performance of the prediction [32]. Considering (iii), the proposed methodology implements a *Model Predictive Controller* (MPC) with *Cross-Entropy Method* (CEM) to online optimize the *impedance control* parameters (i.e, stiffness and damping parameters). The objective of such optimization is the minimization of the human forces applied to the robot in the collaborative application (i.e, the human effort). The proposed control methodology exploits the human-robot interaction dynamics modeling in (ii), performing interaction states predictions over a finite prediction horizon H. A cost function minimizing the human interaction forces (i.e, minimizing the human effort) is defined and exploited in the optimization procedure for the online control parameters optimization.

The here proposed MBRL has been developed in order to identify the human-robot interaction dynamic model, to be exploited in the design and optimization of the presented controller. Other techniques can also be investigated to compute the optimal control action. Dynamic programming algorithms [44–46] can be applied to estimate the target cost function (i.e, approximating it since the theoretical solution of the used Bellman equation is very difficult to be obtained, except for simple systems), e.g., exploiting ANNs. The optimal control signal can then be determined by minimizing the cost function (or maximizing a reward function). However, the use of such techniques does not allow to identify the human-robot interaction dynamics and they are very sensitive to the target cost function to be estimated. Identifying the human-robot interaction dynamics it is instead possible to easily adapt the objective of the controller on the basis of the specific task/scenario. In such a way, it is possible to avoid to perform from scratch the estimation of a new target cost function due to a (partially) new scenario or controller objective.

The proposed MBRL variable impedance controller has been validated in experimental tasks involving a KUKA LBR iiwa 14 R820 as a test platform (Fig. 1). A lifting task of an unknown payload (10 kg) has been used as a reference task. Qualitative (i.e, questionnaire on perceived collaboration) performance have been evaluated, comparing the obtained results with previous developed offline model-free optimized controllers [47] and with the robot manual guidance controller. The proposed MBRL variable impedance controller shows improved human-robot collaboration, empowering the human operator in executing the task. The novelties of the proposed approach are, therefore, related to:

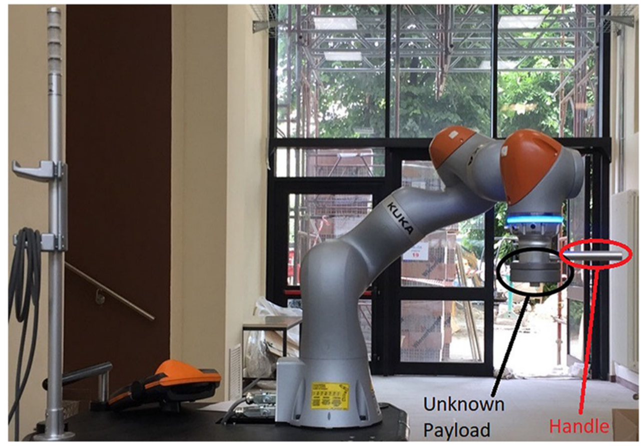


Fig. 1 KUKA LBR iiwa 14 R820 robot involved in the validation setup

- the efficient human-robot interaction dynamics modeling (requiring a limited number of initial training experiments, specifically, 30 initial trials) performed exploiting the proposed ensemble of ANNs;
- the possibility to keep the human-robot interaction dynamic model updated while human-robot collaboration is performed (i.e, accounting for the adaptation of human motor system);
- the online exploitation of the human-robot interaction dynamic model to optimize the impedance control parameters (minimizing the human forces applied to the robot, i.e, to reduce the human effort) by means of the proposed MPC.

The main advantages of the proposed approach are, therefore, related to:

- the possibility to identify the complex human-robot interaction dynamics, accounting for uncertainties in the human-robot collaboration and updating such model to the changes in the human motor system;
- the possibility to exploit the learned complex human-robot interaction dynamics to optimize the control action;
- the possibility to easily modify/adapt the objective of the controller optimization to a different target performance;
- the possibility to exploit all the acquired data (to identify and update the human-robot interaction dynamics) for the control design and optimization without performing a re-identification procedure of the new cost function from scratch (in the case that the objective of the controller changes).

Considering the previously developed offline model-free optimization methodology, the proposed approach is less time-consuming, not requiring a complex learning process and having the possibilities to model varying human-robot interaction dynamics through the human-robot collaboration application (i.e, different human operators, changing fatigue levels, different applications).

General Notation

$\mathbf{M}_r, \mathbf{C}_r, \mathbf{K}_r$: mass, damping, and stiffness impedance matrices, respectively, composed by both the translational and rotational components

$\Delta\mathbf{x}_r$: $\Delta\mathbf{x}_r = \mathbf{x}_r - \mathbf{x}_r^d$, where $\mathbf{x}_r^d$ is the impedance control setpoint and $\mathbf{x}_r$ the robot Cartesian position/orientation vector

$\mathbf{f}_r$: external wrench vector (i.e, force and torque) applied to the manipulator

dt : sampling time for both impedance control setpoint and measurements

$\mathbf{g_d}$: gravity vector $\mathbf{g}_d = [0, 0, 9.81, 0, 0, 0]^T$

$\mathbf{x}_t$: human-robot interaction dynamics state vector at time t; $\mathbf{x}_t = [\mathbf{f}_r, \dot{\mathbf{f}}_r, \mathbf{x}_r, \dot{\mathbf{x}}_r]^T$

D_{rand} : recorded data for the initial ensemble of ANNs training modeling the human-robot interaction dynamics

D_{RL} : online recorded data for the update of the human-robot interaction dynamic model exploiting the ensemble of ANNs

$\mathbf{u}_t$: MPC control action vector at time t; $\mathbf{u}_t = [diag(\mathbf{K}_r), diag(\mathbf{C}_r)]^T$

H : MPC finite prediction horizon (i.e, MPC optimization sampling time);

τ : torque vector output from impedance control

2 Methodology

The main purpose of this paper is to design a *model-based reinforcement learning* (MBRL) variable impedance control schema (a) capable to model the complex human-robot interaction dynamics, (b) online optimizing the impedance parameters (c) to empower/assist (i.e, minimizing human forces applied to the robot, i.e, minimizing the human effort) the human operator in collaborative onerous tasks. To do that, (i) an impedance controller to achieve a compliant robot behavior, (ii) a human-robot interaction dynamics modeling approach (offline trained and online updated) relying on an ensemble of Artificial Neural Networks, and (iii) a Model Predictive Controller employing a Cross-Entropy Method for the online optimization of the impedance parameters (i.e, stiffness and damping parameters) are proposed. The MBRL variable impedance control schema is shown in Fig. 2.

In the following, the implemented impedance controller (Section 2.1), the ensemble of ANNs approach (Section 2.2.2) and the MPC with CEM (Section 2.3) are detailed.

2.1 Cartesian Impedance Control

The inner Cartesian impedance control [48] is considered to implement the proposed MBRL variable impedance controller:

$$\mathbf{M}_r\,\ddot{\mathbf{x}}_r + \mathbf{C}_r\,\dot{\mathbf{x}}_r + \mathbf{K}_r\,\Delta\mathbf{x}_r = \mathbf{f}_r. \tag{1}$$

Impedance control receives as an input the interaction state $\mathbf{x}_t$ and the impedance control stiffness and damping matrices. The updated impedance control setpoint $\mathbf{x}_r^d$ is applied at each control-step on the basis of the following law:

$$\mathbf{x}_r^d = \mathbf{x}_r + \mathbf{K}_r^{-1}\mathbf{f}_r + \widehat{M}_p\mathbf{K}_r^{-1}\mathbf{g}_d. \tag{2}$$

The term $\mathbf{K}_r^{-1}\mathbf{f}_r$ allows deforming the impedance control setpoint $\mathbf{x}_r^d$ to assist the operator exploiting the measured force, while the term $\widehat{M}_p\mathbf{K}_r^{-1}\mathbf{g}_d$ allows to compensate for the estimated payload weight $\widehat{M}_p$. Indeed, the payload

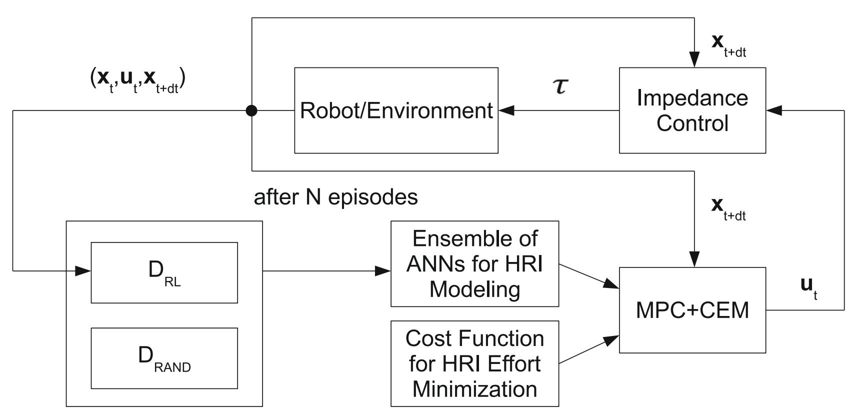


Fig. 2 Model-based reinforcement learning variable impedance control schema. The ensemble of ANNs employed for the human-robot interaction dynamics modeling (offline trained and online updated), the Model Predictive Controller employing the Cross-Entropy Method for optimization of control parameters, and impedance controller are highlighted

weight $\widehat{M}_p$ is estimated on the basis of the measured forces before the collaborative task starts (i.e, in this task-phase no interaction between the human and the robot is present). The forces are measured in a 1 s time-window, the vertical force component is computed and averaged, so that $\widehat{M}_p$ is estimated as:

$$\widehat{M}_p = \frac{\frac{\sum_{i=0}^{n_{sample}} f_{r,z}(i)}{n_{sample}}}{g_{d,z}}, \tag{3}$$

where n_{sample} is the number of samples in the considered time-window. The $\widehat{M}_p \mathbf{g}_d$ estimation is also used to remove the payload offset from force measurements $\mathbf{f}_r$ during the next human-robot collaboration phase.

The stiffness and damping parameters are online optimized and updated to the impedance controller by the MPC, exploiting the human-robot interaction dynamics modeling performed by the ensemble of ANNs. In the following, the human-robot interaction dynamic model learning methodology based on an ensemble of ANNs, and the MPC employing the CEM are detailed.

Remark 1 It has to be underlined that the KUKA LBR iiwa 14 R820 impedance control damping is imposed as damping ratio diagonal matrix $\mathbf{h}_r$ ($\mathbf{C}_r = 2\sqrt{\mathbf{M}_r\mathbf{K}_r}\mathbf{h}_r$).

Remark 2 It has to be remarked that the estimation of the payload weight doesn't need to be accurate. As it will be shown in Section 2.2.2, the ensemble of ANNs will be able to capture human-robot interaction dynamics uncertainties also related to the payload weight.

Remark 3 It has to be underlined that the proposed paper considers translational degrees of freedom (DoFs), while involving pre-defined rotations of robot end-effector (fixed rotations are considered for the reference task described in Section 3). Such an assumption is common in many real tasks (such as insertions, assemblies, etc.) where no re-orientation of the robot end-effector is required, or when a bulky and/or heavy payload is manipulated and rotation path is pre-calculated (i.e, defining a reference trajectory for the robot) [9, 49] (see https://www.youtube.com/watch?v=mdYpZZw93YE). The extension to rotations is under development, on the basis of the quaternion description.

2.2 Human-Robot Interaction Dynamics Modeling

2.2.1 Model-Based Reinforcement Learning vs. Model-Free Reinforcement Learning

Reinforcement learning [50] is one of the most effective methodologies to perform the mapping (i.e, to identify the correlation) between inputs and outputs of a dynamic system for optimization purposes. Reinforcement learning algorithms can be broadly classified in two main families: model-free and model-based methodologies [51]. Considering model-free reinforcement learning approaches, the policy is found without the need to estimate a model of the system dynamics, while model-based reinforcement learning approaches build a transition dynamics model of the system dynamics, used for internal simulations (to improve the policy) and predictions of the rewards and optimal actions. While being more computationally intensive w.r.t. model-free methodologies, the model-based reinforcement learning is more data-efficient. In fact, in robotics, this characteristic significantly reduces the physical experiments to be performed (i.e, model-based approaches are less time-consuming). In addition, one of the main advantages of learning the transition dynamics is that by changing the reward or cost function that is used for evaluation of the actions, new behaviors and objectives can be achieved without the need to learn from scratch which is the case for other methods. [52–54]. On the other hand, transition dynamics models have a huge influence on the performance of model-based reinforcement learning algorithms. In fact, the policy learning relies on the accuracy of their predictions. In Section 2.2.2 the proposed approach to learn the transition dynamics model (i.e, the human-robot interaction dynamics) to be used in the proposed model-based reinforcement learning approach is described.

2.2.2 Ensemble of Artificial Neural Networks Modeling the Human-Robot Interaction Dynamics

In order to design the proposed model-based reinforcement learning approach, human-robot interaction dynamics (i.e, the transition dynamics) has to be modeled. Human-robot interaction dynamics has a very complex nonlinear behavior, difficult to be modeled [39]. Commonly, works in the human-robot collaboration field define simplified human-robot interaction dynamics models (e.g., impedance models) to be used for the model-based control design [55]. However, such simplified models are not able to account for the real complexity of such dynamics, limiting the controller performance. Therefore, there is the need to model the complete nonlinear human-robot interaction dynamics to improve the performance of the collaboration.

There are many ways to learn the transition dynamics model required by the model-based reinforcement learning approach [52]. Artificial Neural Networks (ANNs) have some advantages over other methods: ANNs are general function approximators capable of learning arbitrary complex nonlinear functions which is often present in robotics. Moreover, ANNs are very efficient and can easily scale-up with the data size thanks to parallelization with modern

GPUs, and ANNs are the only function approximators that can work with high dimensional state spaces. However, because of their expressivity, ANNs tend to overfit in low data regime. To counter-act such issue, it is shown in [32, 43, 56] that using an ensemble of ANNs is an effective solution to capture model uncertainty where there are less data. In the region of the state-space where there are more data availability, ANNs agree with each other. In the region where there are less data, ANNs tend to disagree with each other, hence, capturing the uncertainty over the models. In other words, each ANN is certain about its predictions but its model cannot be considered reliable. Therefore, to capture the dynamics uncertainties, a model describing the human-robot interaction dynamics can be created exploiting the ensemble of ANNs (Fig. 3), to be used by the MPC controller for control action $\mathbf{u}_t$ optimization.

The following procedure has been implemented in order to train and update the ensemble of ANNs:

- **defining the ANN architecture**: There is no standard way of finding a good ANN architecture, it is an iterative procedure. The intuition is to have enough layers and hidden units to capture the transition dynamics. In [32] it has been shown that the architecture with 5 hidden layers and 512 hidden units is able to capture the dynamics of wide variety of tasks. Hence, this architecture is used in this work.
- **defining the input layer**: Learning the transition dynamics is performed defining the input layer including both the states of the human-robot system $\boldsymbol{x_t}$ and the control inputs $\boldsymbol{u_t}$ at time t (Fig. 3).
- **defining the output layer**: Learning the transition dynamics can be difficult when the states $\mathbf{x}_t$ and $\mathbf{x}_{t+dt}$ are too similar and the control input has seemingly little effect on the output. To overcome this issue, in [53] it is proposed instead to learn a function of the target dynamics $\widehat{\mathbf{f}}_w(\mathbf{x}_t, \mathbf{u}_t)$ that predicts the change in the state $\mathbf{x}_t$ over the time step duration dt (Fig. 3):

$$\widehat{\mathbf{x}}_{t+dt} = \mathbf{x}_t + \widehat{\mathbf{f}}_w(\mathbf{x}_t, \mathbf{u}_t). \tag{4}$$

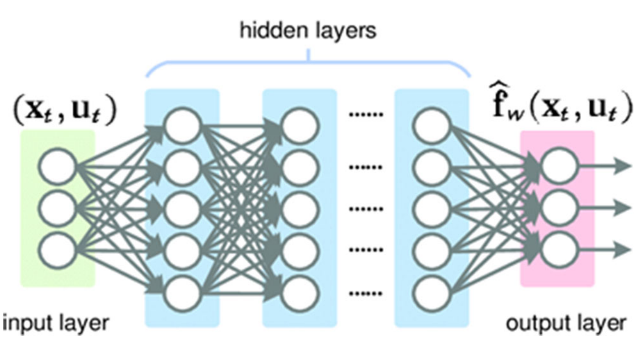


Fig. 3 Typical architecture of the ANN providing a model of the variation of the human-robot interaction dynamics $\widehat{\mathbf{f}}_w(\mathbf{x}_t, \mathbf{u}_t)$.

- **hidden layers**: Deep learning models consist of more than one hidden layer. In order to learn any complex function, such models require to have a nonlinear transformation of input data which is done by hidden layers. The ReLU (Rectified Linear Unit) function can be used for this purpose [57].
- **initial data set for training**: For initial training of ANNs, the target task is executed using random control input parameters and recorded the output data during the human-robot collaboration (D_{rand}). Then, the data is put in the correct format (i.e, with respect to Section 2.2.2).
- **defining the ensemble size**: The size of the ensemble of ANNs is selected to be the minimum size required to capture model uncertainties. Such a procedure is iterative. From experimental tests, a size of 5 ANNs has been found to be a good solution for the proposed problem.
- **ANNs periodic update**: The ensemble of ANNs is updated periodically (i.e, accounting for the adaptation of the human motor system) using D_{RL} data after 3 new task executions, improving and adapting the model

Remark 4 Data D_{rand} and D_{RL} for initial training and online update are recorded if a human force is detected by the robot (in particular, if $|\ \mathbf{f}_r\ |> f_{int}$, with $f_{int} = 3$ N) in order to avoid the recording of not meaningful data.

2.3 Online Optimization of Impedance Control Parameters

2.3.1 Cost Function Design

The objective of the MPC is to optimize the impedance control parameters (i.e, stiffness and damping) in order to minimize the interaction forces during the human-robot collaboration (i.e, minimizing human effort). Indeed, to achieve the goal, the following quadratic cost function $C(\mathbf{x}_t, \mathbf{u}_t)$ (a function of the interaction state $\mathbf{x}_t$ and control action $\mathbf{u}_t$) is proposed:

$$C(\mathbf{x}_t, \mathbf{u}_t) = \mathbf{x}_t^T \mathbf{Q} \mathbf{x}_t + \mathbf{q} \mathbf{x}_t + \mathbf{u}_t^T \mathbf{R} \mathbf{u}_t + \mathbf{r} \mathbf{u}_t, \tag{5}$$

where $\mathbf{Q}$ and $\mathbf{R}$ are the optimization gain matrices, $\mathbf{q}$ and $\mathbf{r}$ are the optimization gain vectors. $\mathbf{Q}$ matrix has non-zero values k_Q only on diagonal elements in correspondence of the wrench states related to Cartesian forces, $\mathbf{q}$ vector has non-zero values k_q only in correspondence of the wrench state related to Cartesian forces, $\mathbf{R}$ is imposed to $\mathbf{R} = \mathbf{0}$, $\mathbf{r}$ vector is imposed to $\mathbf{r} = \mathbf{0}$.

The cost function in Eq. 5 is then used in the MPC to optimized the impedance control parameters (i.e, to optimize the control action $\mathbf{u}_t$).

Remark 5 Having the above definition of optimization gain matrices/vectors means that only interaction Cartesian forces are taken into account in the optimization. Therefore, Cartesian force derivative $\dot{\mathbf{f}}_r$, position $\mathbf{x}_r$ and velocity $\dot{\mathbf{x}}_r$ do not have any influence on the optimization.

2.3.2 Model Predictive Control with Cross-Entropy Method

The learned interaction dynamics $\widehat{\mathbf{f}}_w(\mathbf{x}_t, \mathbf{u}_t)$ Eq. 4, together with the cost function $C(\mathbf{x}_t, \mathbf{u}_t)$ Eq. 5, are employed by an MPC in order to optimize the impedance control parameters (i.e, the control action $\mathbf{u}_t$). Defining the MPC finite prediction horizon H, a sequence of control actions $\mathbf{U}_t^{(H)} = (\mathbf{u}_t, ..., \mathbf{u}_{t+H-dt})$ is optimized, using the learned dynamics models to predict future states:

$$\mathbf{U}_t^{(H)} = \underset{\mathbf{U}_t^{(H)}}{\arg\min} \sum_{\xi=t}^{t+H-dt} C(\widehat{\mathbf{x}}_\xi, \mathbf{u}_\xi); \widehat{\mathbf{x}}_t = \mathbf{x}_t, \widehat{\mathbf{x}}_{\xi+dt} = \widehat{\mathbf{x}}_\xi + \widehat{\mathbf{f}}_w(\mathbf{x}_\xi, \mathbf{u}_\xi). \tag{6}$$

The exact calculation of the optimum of Eq. 6 is not possible due to the high nonlinearity of the problem. In order to calculate an approximate solution of the above finite-horizon control problem, the Cross-Entropy Method is exploited [58]. The CEM is a probabilistic optimization belonging to the field of Stochastic Optimization. The strategy of the algorithm is to sample the problem space and approximate the distribution of good solutions. This is achieved by assuming a distribution of the problem space (such as Gaussian), and sampling from this distribution to generate candidate solutions. Then, the distribution is updated on the basis of the best discovered candidate solutions. As the algorithm progresses, the distribution becomes more refined, until it focuses on the area of optimal solutions. Pseudocode of CEM is presented in Algorithm 1.

The procedure for calculating the best control action $\mathbf{u}_t$ is repeated until convergence to optimal values. Furthermore, rather than have the controller executing the control action sequence in open-loop, an MPC has been implemented [59]. The controller executes the control action $\mathbf{u}_t$, receives updated state information $\mathbf{x}_{t+dt}$, and recalculates the optimal control action sequence at the next time step. The pseudocode describing its implementation is given in Algorithm 2.

Algorithm 1 Pseudocode for Cross-Entropy Method. $Learn_{rate} = \alpha \in [0, 1]$ is considered, with typical value of 0.7.

```
Input: Problem_size, Samples_num,
       UpdateSamples_num, Learn_rate, Var_min;
Output: S_best;
Means ← InitializeMeans();
Variances ← InitializeVar();
S_best ← 0;
while Max(Var) ≤ Var_min do
    Samples ← 0;
    for i = 0 to Samples_num do
        Samples ← GenerateSample(Means, Var);
    end
    EvaluateSamples(Samples);
    SortSamplesByQuality(Samples);
    if Cost(Sample_s0) ≤ Cost(S_best) then
        S_best ← Samples_s0;
    end
    Samples_selected ← SelectBestSamples(Samples,
    UpdateSamples_num);
    for i = 0 to Problem_size do
        Means_i ← Means_i + Learn_rate ×
        Mean(Samples_selected, i);
        Var_i ←
        Var_i + Learn_rate × Var(Samples_selected, i);
    end
end
return S_best;
```

Remark 6 It has to be underlined that the optimization of the control action $\mathbf{u}_t$ and its update is performed when the human force is detected, i.e, when the interaction force $|\mathbf{f}_r| > f_{int}$, with $f_{int} = 3$ N.

Remark 7 It has to be underlined that constraints can be included in the MPC (such as for safety rules implementation) in order to impose limitations to the control action optimization.

Algorithm 2 Pseudocode for MPC.

```
gather dataset D_Rand of random trajectories;
initialize empty dataset D_RL;
randomly initialize ensemble of N, f̂_w neural networks;
for iter=1 to max-iter do
    train all f̂_w(x_t, u_t) functions using D_Rand, and
    D_RL;
    for ξ = t to H do
        get current states x_ξ;
        use f̂_w(x_ξ, u_ξ) to estimate optimal control
        action sequence U_t^(H) (Algorithm 1);
        execute first control action u_t from U_t^(H);
        get the new state x_{t+dt} ;
        add (x_t, u_t, x_{t+dt} - x_t) to D_RL;
    end
end
```

3 Experimental Evaluation in Human-Robot Collaboration Task

3.1 Validation Scenario

The considered scenario for the validation of the here proposed MBRL variable impedance controller is the same as in the work described in [47].

10 healthy males subjects and 5 healthy females subjects, for a total of 15 healthy subjects (age 29 $\pm$ 4 years), have performed the validation experiments.

A KUKA LBR iiwa 14 R820 robot has been involved in the experimental tests to collaborate with the human (Fig. 1). The human-robot interaction forces have been measured using the joint torque sensors of the manipulator. The BTS FreeEMG300 system has been used to measure the upper-limb electromyography activity of the subject involved in the experiments.

A lifting task of an unknown (to the robot controller) payload (10 kg) has been considered as a reference task (Fig. 4). The complete task description can be found in [47]. The MBRL variable impedance control has been applied to the vertical Cartesian DoF z, while x and y components of Eq. 2 have been kept constant. In such a way, authors would like to highlight the evaluation of the perceived human-robot collaboration in the most critical Cartesian direction (i.e, in which the payload has to be compensated). The application

Fig. 4 Cooperative lifting task highlighting for the evaluation of the proposed MBRL variable impedance controller

of the proposed MBRL variable impedance controller to the x and y Cartesian DoFs is straightforward. The vertical trajectory performed by the human in interaction with the robot has been online calculated by Eq. 2 as described in Section 2.1, while the related impedance control parameters have been online optimized by the MPC as described in Section 2.3. The end-effector orientation has been kept fixed during the task execution, as stated in Section 2.1. As described in Section 2.1, before to start the human-robot collaboration task (i.e, without any interaction between the human and the robot), a quick estimation of the payload weight $\widehat{M}_p$ is performed, exploiting the measured forces at the robot end-effector, so that its weight can be compensated in Eq. 2.

Performance of the proposed MBRL variable impedance controller have been compared with previously developed offline model-free optimized controllers [47]. Additionally, the proposed MBRL variable impedance controller has been compared with the KUKA iiwa 14 R820 high-performance manual guidance controller with perfect-weight compensation and uncertain-weight compensation (30% of difference between the real weight and the compensated weight), typical CAD part weight modeling errors [47]. Thus, six controllers have been implemented, tested and compared: (1) an interaction forces-based controller iF_{ctrl} [47], (2) a trajectory smoothness-based controller iJN_{ctrl} [47], (3) an electromyography-based controller $iEMG_{ctrl}$ [47], (4) an MBRL variable impedance controller $MBRL_{ctrl}$ (Section Section 2), (5) a perfect-weight compensation KUKA iiwa manual guidance controller $KUKA_{ctrl}$, (6) an uncertain-weight compensation KUKA iiwa manual guidance controller $KUKA_{ctrl,u}$. To avoid any influence on the results related to the sequence of the controllers, their testing order has been randomized and no information of the current tested strategy has been given to the subjects [47].

Remark 8 The proposed MBRL impedance controller can be applied to industrial manipulators that are not equipped with joint torque sensors. In fact, adopting a force/torque sensor mounted at the robot end-effector (between the robot flange and the human-robot interaction handle to being able to measure the interaction forces/torques applied by the subject) or exploiting interaction force estimation based on motor currents measurements [60], it is possible to apply the proposed methodology without any modification.

3.2 Software

The proposed MBRL variable impedance controller has been implemented on the KUKA LBR iiwa 14 R820 on the basis of the ROS framework, exploiting the *iiwa_stack* [61] (Fig. 5). The ensemble of ANNs has been implemented on

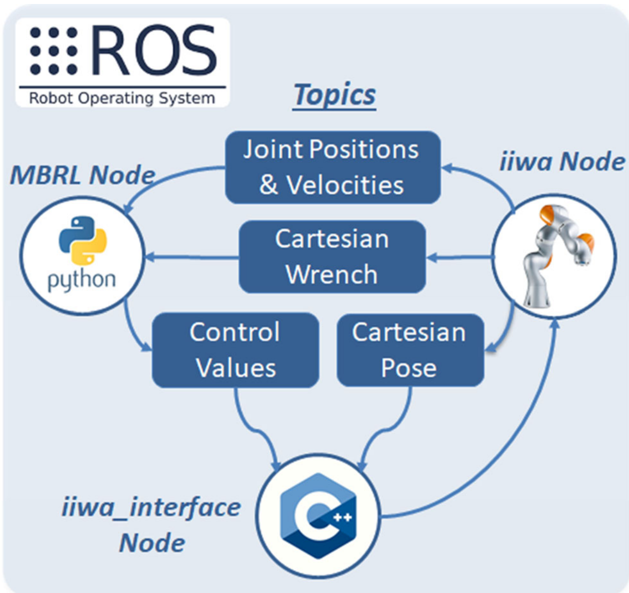


Fig. 5 Proposed MBRL variable impedance controller implemented in the ROS framework

the basis of the *PyTorch* library [62]. CEM and MPC have been *ad hoc* implemented in *Python*. A c++ wrapper has been developed in order to have ROS nodes communicating with the control layer.

A control frequency of 100 Hz (i.e, $dt = 0.01$ s) has been imposed to update the impedance control setpoint. Such control frequency has been the maximum achievable reliable control frequency by means of the iiwa_stack. In fact, iiwa_stack for Cartesian impedance control exploits the KUKA iiwa Sunrise *DirectServo* motion commands, allowing to control the robot between 50 and 100 Hz. The same frequency has been used to sampling the robot states (i.e, interaction force $\mathbf{f}_r$, derivative of the interaction force $\dot{\mathbf{f}}_r$, Cartesian end-effector position $\mathbf{x}_r$ and velocity $\dot{\mathbf{x}}_r$).

The ensemble of ANNs (Section 2.2.2) internal parameters [43] have been tuned on the basis of experimental validation and [32] to the values shown in Table 1. 30 initial sets of impedance control parameters have been defined to train the ensemble of ANNs. In such a training phase, stiffness has been imposed equal to [500, 1000, 2000, 3000, 4000, 5000] N/m, while damping ratio has imposed equal to [0.1, 0.3, 0.5, 0.7, 0.9]. All the possible combinations have been used for the initial training. The CEM parameters (Section 2.3) internal parameters [58] have been tuned to the values shown in Table 2. The

Table 1 ANNs internal parameters setting

Ensemble Size	5
Number of hidden layers	5
Number of hidden units	512

Table 2 MPC and CEM internal parameters setting

Number of samples	64
Number of elites	32
Learn Rate	0.9
Maximum Iterations	10

setting of such parameters have been performed to achieve a computation time less than 0.2 s (i.e, the MPC finite prediction horizon $H = 0.2$ s). In such a way, the MPC control frequency has been imposed to 5 Hz (i.e, for stiffness and damping parameters update). In fact, in order to guarantee stability in the human-robot collaboration [63] and due to the slow dynamics displayed by both impedance control and human's arm [64], such control frequency has been selected to avoid any drastic change in the coupled human-robot interaction dynamics, while being an order of magnitude faster than the common human motion dynamics ($\simeq 0.5$ Hz [65]). The gain values k_Q related to the matrix $\mathbf{Q}$ in Eq. 5 has been imposed equal to 5. The cost function values k_q related to the vector $\mathbf{q}$ in Eq. 5 has been imposed equal to 5.

The MPC updates online the impedance control stiffness and damping ratio parameters along the vertical axis z on the basis of the optimized values in the range [500, 5000] N/m and [0.1, 0.9] for stiffness and damping ratio, respectively. Impedance control stiffness and damping ratio parameters along x and y Cartesian DoFs have been imposed to 2500 N/m and 0.7, respectively. The impedance control rotational stiffness and damping ratio parameters have been imposed as follows: $\mathbf{K}_{r,transl} = diag(300, 300, 300)$ Nm/rad, $\mathbf{h}_{r,transl} = diag(0.9, 0.9, 0.9)$ (i.e, high stiffness and high damping to limit as much as possible deformations and oscillations).

3.3 Experimental Results

3.3.1 Human-Robot Interaction Dynamic Model Validation

The learned human-robot interaction dynamic model is fundamental for the effectiveness of the proposed MBRL variable impedance controller. In fact, on the basis of the learned model, the MPC computes the optimized impedance control parameters to be applied during the human-robot collaboration task. Therefore, such model affects the task dynamics and the human perception of the collaboration with the robot. It is therefore mandatory that the proposed approach is capable to model the complex human-robot interaction dynamics to succeed in the MBRL variable impedance control purposes.

In order to validate the proposed model, Fig. 6 shows the training loss and the test loss for the 5 ANN. ANNs has been trained on the basis of the initial 30 trials, while the

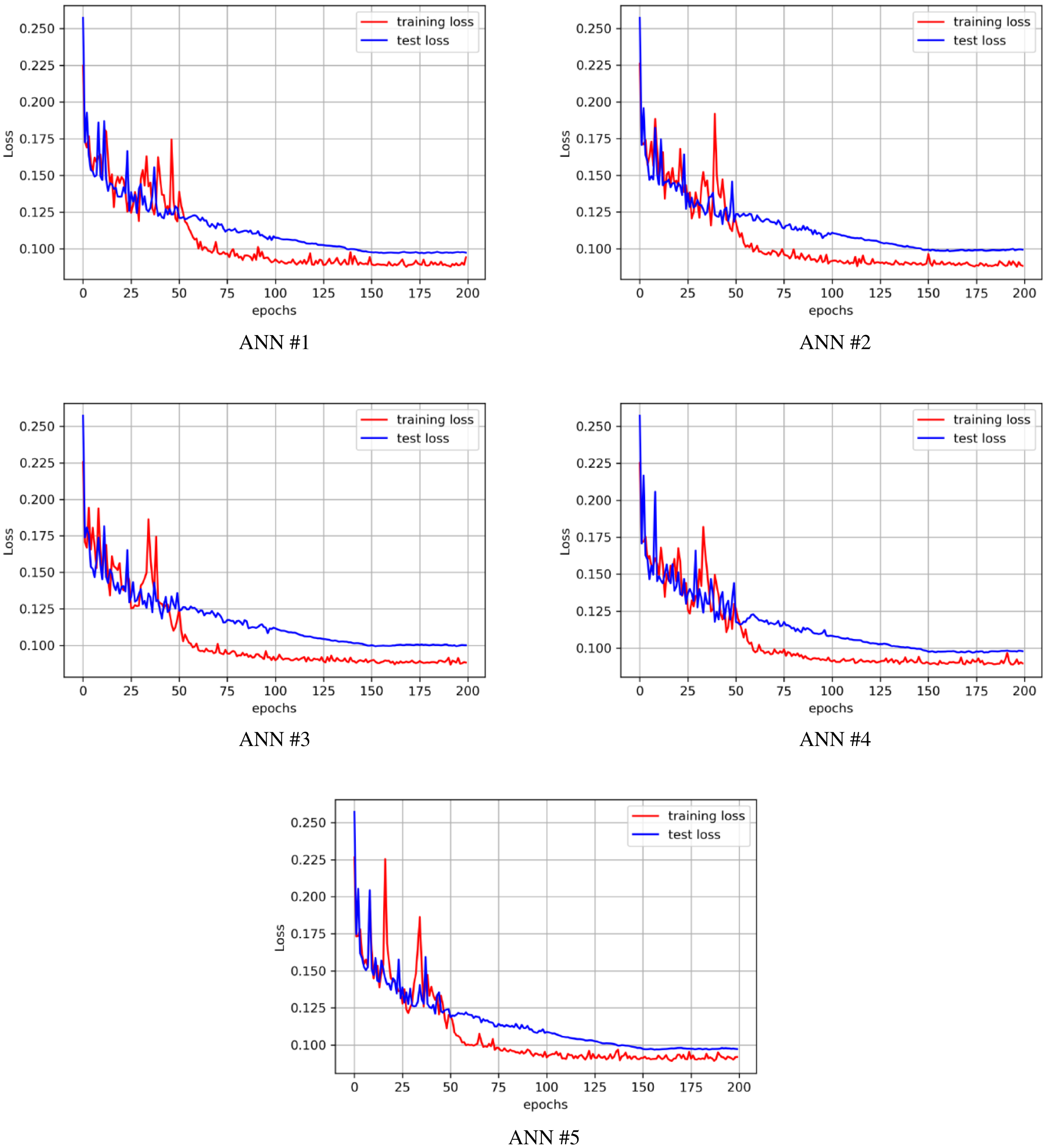


Fig. 6 Training loss (red line) and test loss (blue line) for the 5 ANN

test has been performed considering additional 15 trails with random generated values for stiffness and damping ratio parameters. Such plots highlight that the proposed ensemble of ANNs is capable to model the human-robot interaction dynamics. Figure 7 shows the measured interaction force $f_{r,z}$ along the vertical direction z, and its predictions performed by the trained ANNs with the initial dataset D_{Rand} for one lifting experiment (comparable results are shown for all the other experiments). The ensemble of ANNs is capable to model the uncertainties related to the human-robot interaction dynamics, being effective for the implementation of the proposed MBRL variable impedance

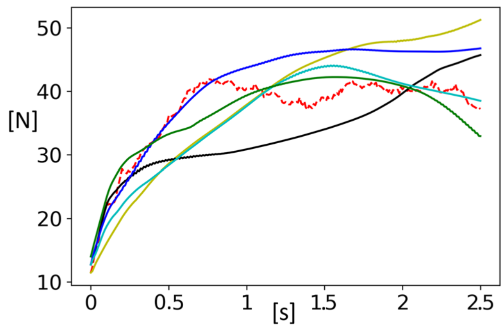


Fig. 7 Measured interaction force $f_{r,z}$ for a lifting task (dashed red line), and the predictions of the five ANNs (continuous lines) are shown

controller. The ensemble of ANNs is then used in Eq. 6 to optimize the impedance control stiffness and damping ration based on the cost function Eq. 5. The human-robot interaction dynamics model is online updated on the basis of the dataset D_{RL} ti improve the model.

3.3.2 Online Impedance Control Parameters Optimization

On the basis of the human-robot interaction dynamic model learned by the ensemble of ANNs, the MPC computes online the optimized impedance control parameters (i.e, stiffness and damping) in order to minimize the human-robot interaction forces. Figure 8 shows the online optimization of the impedance control stiffness $K_{r,z}^{opt}$ in a lifting task experiment. As stated in Section 2.3.2, the optimization of the impedance control parameters is performed when the human force is sensed by the manipulator. As it can be seen from Fig. 8, stiffness values are optimized to small values (i.e, the robot is soft and easy to be positioned while collaborating with the human - results are consistent with optimization in [47]). Picks shown in the Figure are related to very small measured interaction forces $f_{r,z}$, which are related to bigger uncertainties in the learned human-robot interaction dynamic model.

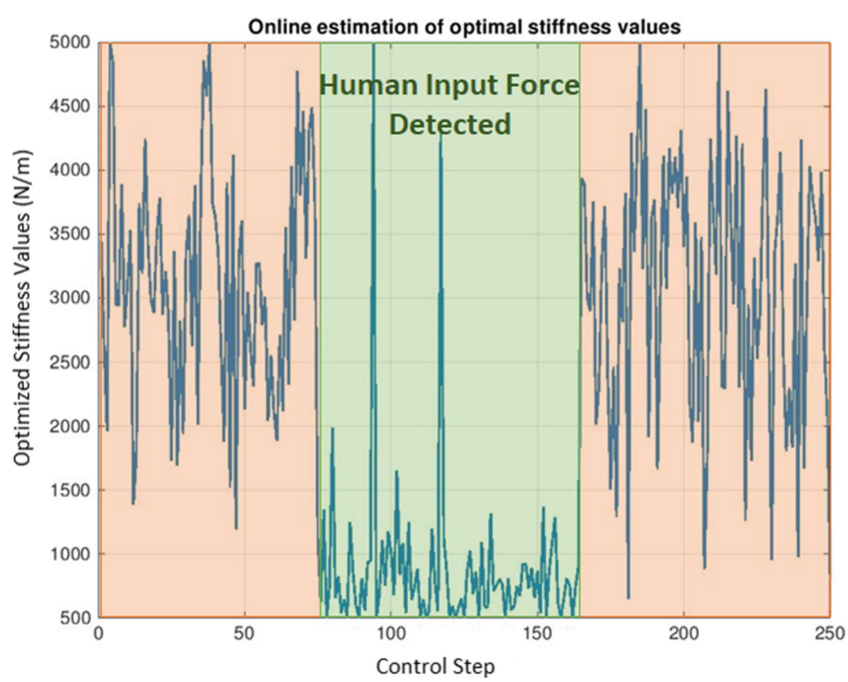
3.3.3 MBRL Variable Impedance Controller Performance in Human-Robot Collaboration

The human-robot collaboration performance have been qualitatively assessed using a questionnaire at the end of each experiment, for all the subjects (as described in [47, 66], defining the following metrics: *Naturalness*, *Smoothness*, *Effort*, *Motion*, *Stability*, *Detection of intention*, *Overall performance*. See [47] for the complete description of the metrics. All the metrics scores are in the range [0, 4] (where 0 is related to very bad performance, 4 to very good performance).

The questionnaires scores of each performance criteria are shown in Fig. 9, while the overall performance criterion is shown in Fig. 10. Each subjects has ranked the proposed criteria w.r.t. each controller proposed in Section 3.1.

Results indicate that the previously developed offline model-free electromyography-based controller and the MBRL variable impedance controller are the most performant methodologies. Their peeformance are improved w.r.t. the perfect-weight compensation KUKA iiwa manual guidance controller $KUKA_{ctrl}$. It is important to underline that the proposed MBRL variable impedance control is

Fig. 8 Optimized impedance control stiffness $K_{r,z}^{opt}$ is shown for a given lifting task



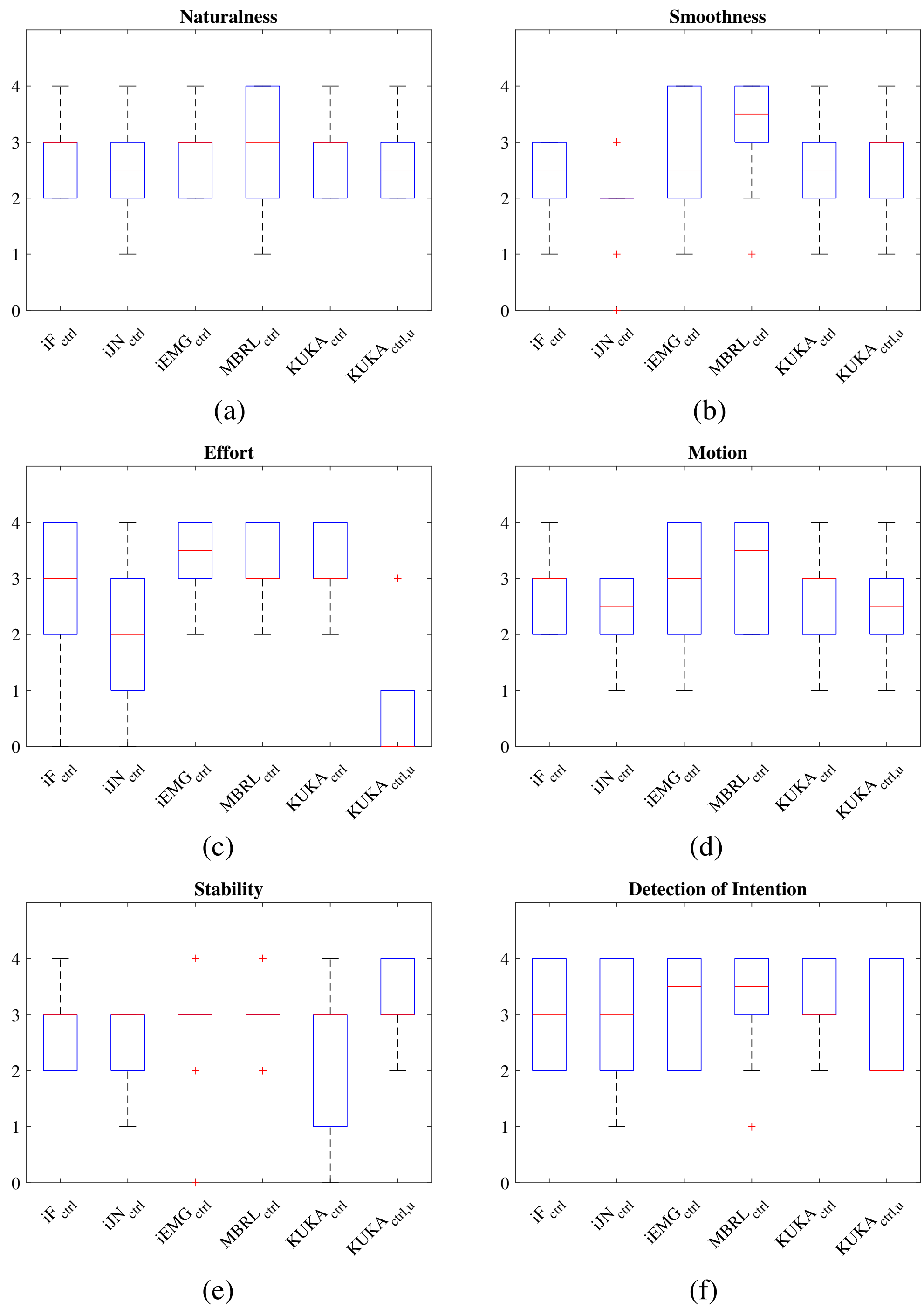


Fig. 9 **a** *Naturalness*, **b** *Smoothness*, **c** *Effort*, **d** *Motion*, **e** *Stability*, **f** *Detection of Intention* criteria of the questionnaire [47] for the different tested controllers

more efficient than the offline model-free iEMG optimized approach, requiring only 30 initial training experiments *vs.* 405 initial training experiments. Moreover, the MBRL variable impedance controller allows to continuously update the learned human-robot interaction dynamic model, being able to capture changes in the interaction dynamics and in the human motor system (not possible in the previously developed approach in [47]). One of the most valuable results is related to the *Effort* criterion. In fact, the *Effort* criterion allows to evaluate the perceived physical load by the human operator (Fig. 9c). In particular, it can be underlined that the uncertain-weight compensation KUKA iiwa manual guidance controller $KUKA_{ctrl,u}$ becomes too heavy for human-robot collaboration even in the presence of a limited weight modeling error. In addition, the proposed MBRL variable impedance controller $MBRL_{ctrl}$ performs better than the perfect-weight compensation KUKA iiwa manual guidance controller $KUKA_{ctrl}$.

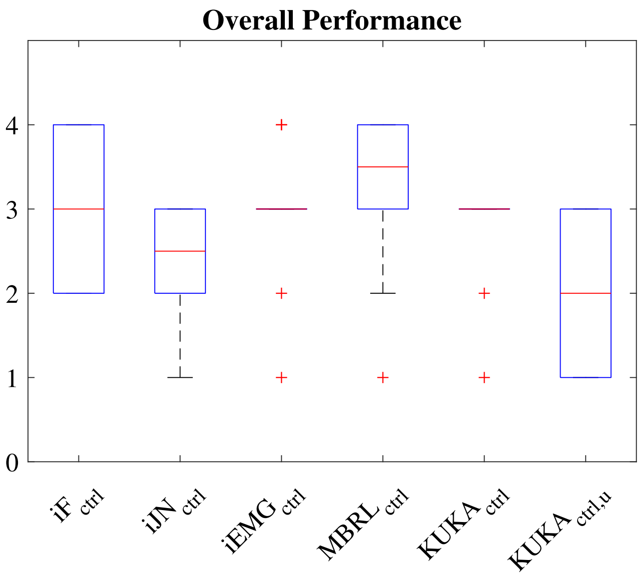


Fig. 10 *Overall performance* criterion of the questionnaire [47] for the different tested controllers

4 Conclusions

The presented work proposes an MBRL variable impedance control to minimize the interaction forces in human-robot collaboration tasks. The proposed approach implements (i) an impedance control, (ii) an ensemble of Artificial Neural Networks, and (iii) a Model Predictive Controller employing a Cross-Entropy Method. The aim of the proposed methodology is to create and online update (i.e, accounting for the adaptation of human motor system) a model of the human-robot interaction dynamics, being exploited for online optimization of impedance control parameters. The proposed approach allows to continuously update the human-robot interaction dynamics, accounting for uncertainties and human motor system adaptations. Optimization is also maintained effective exploiting such model update during the human-robot collaboration. In addition, it allows to easily adapt the objective of the control design and optimization without losing data (relying on the identified and updated human-robot interaction model). A lifting collaborative task has been defined as reference task, involving 15 human subjects. Results are compared with previous developed offline model-free optimized controllers and with the robot manual guidance controller, highlighting better performance for the proposed MBRL variable impedance controller.

To improve the here presented methodology, redundancy optimization is under investigation. The optimization of the impedance setpoint control law is also under consideration, together with the improvement of the cost function definition. In addition, safety rules will be embedded in the MPC controller to ensure safe and reliable interaction with the human. This means that constraints representing safety limitations will be included (such as in [67–69]) in the MPC to guarantee the computation of the optimal control action satisfying both performance and safety. As shown from experimental results, the online use of electromyography measurements can even improve the human-robot collaboration. The online use of such measurements is therefore under consideration to be included in the MPC. Finally, the automatic optimization of the ANNs and MPC parameters based on AI techniques is under investigation.

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References

1. Lasi, H., Fettke, P., Kemper, H.G., et al.: Industry 4.0. Business & Information Systems Engineering **6**(4), 239–242 (2014)
2. Longo, F., Nicoletti, L., Padovano, A.: Smart operators in industry 4.0: a human-centered approach to enhance operators' capabilities and competencies within the new smart factory context. Comput. Indust. Eng. **113**, 144–159 (2017)
3. Vysocky, A., Novak, P.: Human-robot collaboration in industry. MM Sci. J. **9**(2), 903–906 (2016)
4. Schmidtler, J., Knott, V., Hölzel, C., et al.: Human centered assistance applications for the working environment of the future. Occupational Ergonomics **12**(3), 83–95 (2015)
5. Munoz, L.: Ergonomics in the industry 4.0: exoskeletons. J. Ergonomics **8**, e176 (2018)
6. Nahavandi, S.: Industry 5.0–a human-centric solution. Sustainability **11**(16), 4371 (2019)
7. Hogan, N.: Impedance control: an approach to manipulation. In: 1984 American Control Conference, pp. 304–313. IEEE (1984)
8. Roveda, L.: A user-intention based adaptive manual guidance with force-tracking capabilities applied to walk-through programming for industrial robots. In: 2018 15th International Conference On Ubiquitous Robots (UR), pp. 369–376. IEEE (2018)
9. Roveda, L., Haghshenas, S., Prini, A., et al.: Fuzzy impedance control for enhancing capabilities of humans in onerous tasks execution. In: 2018 15th International Conference on Ubiquitous Robots (UR), pp. 406–411. IEEE (2018)
10. Khan, S.G., Herrmann, G., Al Grafi, M., et al.: Compliance control and human–robot interaction: part 1–survey. Int. J. Humanoid Robotics **11**(03), 1430001 (2014)
11. Liang, P., Yang, C., Wang, N., et al.: Implementation and test of human-operated and human-like adaptive impedance controls on baxter robot. In: Conference Towards Autonomous Robotic Systems, pp. 109–119. Springer (2014)
12. Yang, C., Zeng, C., Fang, C., et al.: A dmps-based framework for robot learning and generalization of humanlike variable impedance skills. IEEE/ASME Transactions on Mechatronics **23**(3), 1193–1203 (2018)
13. Grafakos, S., Dimeas, F., Aspragathos, N.: Variable admittance control in pHRI using EMG-based arm muscles co-activation. In: 2016 IEEE International Conference on Systems, Man, and Cybernetics, SMC 2016 - Conference Proceedings, pp. 1900–1905 (2016)
14. Ranatunga, I., Lewis, F., Popa, D.O., Tousif, S.M.: Adaptive admittance control for human-robot interaction using model

reference design and adaptive inverse filtering. IEEE Trans. Control Syst. Technol. **25**(1), 278–285 (2016)
15. Wai, H., Hlaing, W., Myo, A., Sin, T.: Variable admittance controller for physical human robot interaction. In: IECON 2017 - 43rd Annual Conference of the IEEE Industrial Electronics Society, pp. 2929–2934 (2017)
16. Landi, C.T., Ferraguti, F., Sabattini, L., et al.: Admittance control parameter adaptation for physical human-robot interaction. In: Proceedings - IEEE International Conference on Robotics and Automation, pp. 2911–2916 (2017)
17. Secchi, C., Stramigioli, S., Fantuzzi, C.: Position drift compensation in port-hamiltonian based telemanipulation. In: International Conference on Intelligent Robots and Systems, IROS 2006, pp. 4211–4216 (2006)
18. Gomes, W.: Role adaptive admittance controller for human-robot co-manipulation. Robotics - HRC/New-L (2018)
19. Gopinathan, S., Mohammadi, P., Steil, J.J.: Improved human-robot interaction : a manipulability based approach improved human-robot interaction. Robotics - HRC/Control-pHRI,Thesis (2018)
20. Li, Z., Huang, B., Ye, Z., et al.: Physical human-robot interaction of a robotic exoskeleton by admittance control. IEEE Trans. Indust. Electron. **0046**(c), 1–11 (2018)
21. Buchli, J., Theodorou, E., Stulp, F., et al.: Variable impedance control-a reinforcement learning approach. Robotics, Science and Systems. 1. Available form: http://www.cs.washington.edu/homes/etheodor/papers/buchli-RSS2010.pdf%5Cnhttp://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.167.7254&rep=rep1&type=pdf (2010)
22. Dimeas, F., Aspragathos, N.: Reinforcement learning of variable admittance control for human-robot co-manipulation. IEEE Int. Conf. Intell. Robots Sys. **2015-Decem**, 1011–1016 (2015)
23. Rozo, L., Calinon, S., Caldwell, D.G., et al.: Learning physical collaborative robot behaviors from human demonstrations. IEEE Transactions on Robotics. **32**(3), 513–527 (2016)
24. Alqaudi, B., Modares, H., Ranatunga, I., et al.: Model reference adaptive impedance control for physical human-robot interaction. Control Theory Technol. **14**(1), 68–82 (2016)
25. Modares, H., Ranatunga, I., Lewis, F.L., et al.: Optimized assistive human – robot interaction using reinforcement learning. Trans. Cybern. **46**(3), 1–13 (2016)
26. Li, Z., Liu, J., Huang, Z., et al.: Adaptive impedance control of human–robot cooperation using reinforcement learning. IEEE Trans. Indust. Electron. **64**(10), 8013–8022 (2017)
27. Rahimi, H.N., Howard, I., Cui, L.: Neural impedance adaption for assistive human-robot interaction. Neurocomputing **290**, 50–59 (2018)
28. Li, C., Zhang, Z., Xia, G., et al.: Efficient force control learning system for industrial robots based on variable impedance control. Sensors **18**(8), 2018 (2539)
29. Argall, B.D., Chernova, S., Veloso, M., et al.: A survey of robot learning from demonstration. Robotics and Autonomous Systems **57**(5), 469–483 (2009)
30. Lee, J.: A survey of robot learning from demonstrations for human-robot collaboration. arXiv:171008789 (2017)
31. Deisenroth, M., Rasmussen, C.E.: Pilco: a model-based and data-efficient approach to policy search. In: Proceedings of the 28th International Conference on machine learning (ICML-11), pp. 465–472 (2011)
32. Chua, K., Calandra, R., McAllister, R., et al.: Deep reinforcement learning in a handful of trials using probabilistic dynamics models. Robotics - HRC/ML Control,Thesis. Available from: arXiv:1805.12114 (2018)
33. Admiraal, M.A., Kusters, M.J., Gielen, S.C.: Modeling kinematics and dynamics of human arm movements. Motor Control. **8**(3), 312–338 (2004)
34. Rosen, J., Perry, J.C., Manning, N., et al.: The human arm kinematics and dynamics during daily activities-toward a 7 dof upper limb powered exoskeleton. In: 12th International Conference on Advanced Robotics, 2005. ICAR'05. Proceedings, pp. 532–539. IEEE (2005)
35. Rahman, M.M., Ikeura, R., Mizutani, K.: Investigation of the impedance characteristic of human arm for development of robots to cooperate with humans. JSME International Journal Series C Mechanical Systems, Machine Elements and Manufacturing **45**(2), 510–518 (2002)
36. Mobasser, F., Hashtrudi-Zaad, K.: A method for online estimation of human arm dynamics. In: 2006 International Conference of the IEEE Engineering in Medicine and Biology Society, pp. 2412–2416. IEEE (2006)
37. Howard, M., Braun, D.J., Vijayakumar, S.: Transferring human impedance behavior to heterogeneous variable impedance actuators. IEEE Transactions on Robotics **29**(4), 847–862 (2013)
38. Medina, J.R., Endo, S., Hirche, S.: Impedance-based gaussian processes for predicting human behavior during physical interaction. In: 2016 IEEE International Conference on Robotics and Automation (ICRA), pp. 3055–3061. IEEE (2016)
39. Noohi, E., Žefran, M., Patton, J.L.: A model for human–human collaborative object manipulation and its application to human–robot interaction. IEEE Trans. Robotics **32**(4), 880–896 (2016)
40. Ghadirzadeh, A., Bütepage, J., Maki, A., et al.: A sensorimotor reinforcement learning framework for physical human-robot interaction. In: 2016 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), pp. 2682–2688. IEEE (2016)
41. Hamaya, M., Matsubara, T., Noda, T., et al.: Learning assistive strategies for exoskeleton robots from user-robot physical interaction. Pattern Recogn. Lett. **99**, 67–76 (2017)
42. Peternel, L., Tsagarakis, N., Caldwell, D., et al.: Robot adaptation to human physical fatigue in human–robot co-manipulation. Autonomous Robots **42**(5), 1011–1021 (2018)
43. Bekey, G.A., Goldberg, K.Y.: Neural Networks in Robotics, vol. 202. Springer, Berlin (2012)
44. Wang, F.Y., Zhang, H., Liu, D.: Adaptive dynamic programming: an introduction. IEEE Comput Intell Magazine **4**(2), 39–47 (2009)
45. Lewis, F.L., Vrabie, D.: Reinforcement learning and adaptive dynamic programming for feedback control. IEEE Circ Sys Magazine **9**(3), 32–50 (2009)
46. Liu, D., Wei, Q., Wang, D., et al.: Adaptive Dynamic Programming with Applications in Optimal Control. Springer, Berlin (2017)
47. Roveda, L., Haghshenas, S., Caimmi, M., et al.: Assisting operators in heavy industrial tasks: on the design of an optimized cooperative impedance fuzzy-controller with embedded safety rules. Frontiers in Robotics and AI **6**, 75 (2019)
48. Siciliano, B., Villani, L. Robot Force Control, 1st edn. Kluwer Academic Publishers, Norwell (2000)
49. Roveda, L., Castaman, N., Ghidoni, S., et al.: Human-robot cooperative interaction control for the installation of heavy and bulky components. In: 2018 IEEE International Conference on Systems, Man, and Cybernetics (SMC), pp. 339–344. IEEE (2018)
50. Nikolaidis, S., Ramakrishnan, R., Gu, K., et al.: Efficient model learning from joint-action demonstrations for human-robot collaborative tasks. In: 2015 10th ACM/IEEE International Conference on Human-Robot Interaction (HRI), pp. 189–196. IEEE (2015)

51. Gläscher, J., Daw, N., Dayan, P., et al.: States versus rewards: dissociable neural prediction error signals underlying model-based and model-free reinforcement learning. Neuron **66**(4), 585–595 (2010)
52. Polydoros, A.S., Nalpantidis, L.: Survey of model-based reinforcement learning: applications on robotics. J. Intell. Robotic Sys. **86**(2), 153–173 (2017)
53. Nagabandi, A., Kahn, G., Fearing, R.S., et al.: Neural network dynamics for model-based deep reinforcement learning with model-free fine-tuning. Robotics - HRC/ML Control, Thesis. Available from: arXiv:1708.02596 (2017)
54. Sutton, R.S., Barto, A.G.: Reinforcement Learning: an Introduction. MIT Press, Cambridge (2018)
55. Campeau-Lecours, A., Otis, M.J., Gosselin, C.: Modeling of physical human–robot interaction: admittance controllers applied to intelligent assist devices with large payload. Int. J. Adv. Robotic Sys. **13**(5), 1729881416658167 (2016)
56. Zhou, Z.H., Wu, J., Tang, W.: Ensembling neural networks: many could be better. Artificial Intelligence **137**(1-2), 239–263 (2002). Available from: http://dx.doi.org/10.1016/j.artint.2010.10.001
57. Srivastava, R.K., Masci, J., Gomez, F., et al.: Understanding locally competitive networks. Available from: arXiv:1410.1165 (2014)
58. Brownlee, J.: Clever algorithms: nature-inspired programming recipes. Jason Brownlee (2011)
59. Findeisen, R., Allgöwer, F., Biegler, L.T.: Assessment and Future Directions of Nonlinear Model Predictive Control. Springer, Berlin (2007)
60. Wahrburg, A., Bös, J., Listmann, K.D., et al.: Motor-current-based estimation of cartesian contact forces and torques for robotic manipulators and its application to force control. IEEE Trans. Automation Sci. Eng. **15**(2), 879–886 (2017)
61. Technical University of Munich, Germany. (2019). https://github.com/ifl-camp/iiwa_stack. Online; Accessed 24 Jul 2019
62. Paszke, A., Gross, S., Chintala, S., et al.: Automatic differentiation in PyTorch. In: NIPS Autodiff Workshop (2017)
63. Keemink, A.Q., van der Kooij, H., Stienen, A.H.: Admittance control for physical human–robot interaction. The Int. J. Robotics Res. **37**(11), 1421–1444 (2018)
64. Mitsantisuk, C., Katsura, S., Ohishi, K.: Kalman-filter-based sensor integration of variable power assist control based on human stiffness estimation. IEEE Trans. Indust. Electron. **56**(10), 3897–3905 (2009)
65. Warrier, R.B., Devasia, S.: Kernel-based human-dynamics inversion for precision robot motion-primitives. In: 2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), pp. 6037–6042. IEEE (2018)
66. Rahman, S.M., Ikeura, R.: Cognition-based control and optimization algorithms for optimizing human-robot interactions in power-assisted object manipulation. J Inf Sci Eng. **32**(5), 1325–1344 (2016)
67. Norouzzadeh, S., Lorenz, T., Hirche, S.: Towards safe physical human-robot interaction: an online optimal control scheme. In: 2012 IEEE RO-MAN: the 21st IEEE International Symposium on Robot and Human Interactive Communication, pp. 503–508. IEEE (2012)
68. Avanzini, G.B., Zanchettin, A.M., Rocco, P.: Constrained model predictive control for mobile robotic manipulators. Robotica **36**(1), 19–38 (2018)
69. Sun, K., Mou, S., Qiu, J., et al.: Adaptive fuzzy control for nontriangular structural stochastic switched nonlinear systems with full state constraints. IEEE Trans. Fuzzy Sys. **27**(8), 1587–1601 (2018)

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